

KRISH JOSHI

Software Engineer | Systems Programming • High-Performance Computing • Backend Infrastructure

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PROFESSIONAL SUMMARY

Software Engineer specializing in **Systems Programming, High-Performance Computing, and Algorithmic Benchmark Architecture** in Python, C/C++, OpenCL, SQL, and Linux. Independent contractor for Handshake (Project Dynamo) designing isolated containerized runtimes, low-level GPU compute kernels, storage recovery engines, and deterministic automated verification suites. Strong foundation in memory safety, OS internals, concurrency (SIMD/SIMT), and discrete algorithms with a dedicated focus on robust, production-grade software delivery.

WORK EXPERIENCE

Software Engineer & Benchmark Developer (Independent Contractor)

July 2026 – September 2026

Handshake — Project Dynamo

Remote / Global

- **Architected & Containerized Multi-Tier Evaluation Environments:** Built 10 hardened, reproducible Docker environments (Harbor benchmark format) evaluating frontier AI models across low-level GPU kernels, cryptographic protocols, distributed DAGs, and Linux storage subsystems.
- **Deterministic Automated Quality & Verification Gates:** Engineered comprehensive Python test suites (PyTest, CTRF schema reporting, SHA-256 assertions) incorporating exact IEEE-754 float tolerances and sandboxed assertions to evaluate complex technical solutions with a 100% pass rate.
- **Runtime Sandboxing & Resource Isolation:** Designed strict file-system sandboxing, resource capping (memory/CPUs), and boundary input sanitization preventing memory leaks, privilege escalation, and unintended host mutation in evaluation runtimes.
- **High-Performance Algorithmic Reference Solvers:** Implemented production-grade reference solutions in Python and C/C++ achieving a 100% verification pass rate across dynamically generated, edge-case constraint inputs.

KEY TECHNICAL PROJECTS (PROJECT DYNAMO)

Quantized GEMM GPU Compute Engine & Bit-Unpacking Kernel

OpenCL C, C/C++, Python, PyTest, SIMT Concurrency

- Engineered a high-throughput **W4A16 matrix multiplication kernel** executing tiled dot products with asymmetric 4-bit packed weights and FP32 accumulators on GPU architectures.
- Eliminated workgroup race conditions using local memory barrier fences and designed boundary-safe bit-shift unpacking logic for dynamic, non-power-of-two matrix shapes.
- Built an automated Python-OpenCL test harness verifying numerical accuracy against FP64 reference baselines across ragged multi-dimensional tensors.

LVM Thin-Pool Storage Extent Carving & Block Recovery Engine

Python, C, Linux Systems, SQLite, Galois LFSR, Block Storage

- Engineered a low-level storage recovery engine to resolve metadata exhaustion deadlocks, repair corrupted LVM thin pool superblocks, and restore binary volume metadata via thin-provisioning tools.
- Reconstructed physical-to-logical block allocation tables and carved fragmented SQLite databases from raw 64KB extents by descrambling Galois LFSR keystream transformations and parsing damaged header offsets.
- Validated database integrity and extent boundaries through automated SHA-256 verification and row-level relational consistency checks.

Distributed Execution Graph & DAG State Reconciliation Engine

Python, Graph Algorithms, SQLite, PyTest, JSONL

- Architected an algorithmic reconciliation engine to parse and reconstruct execution paths from 3,500+ line branching cluster logs, resolving node rollbacks and stateful resume graphs.
- Implemented recursive tree pruning, cycle detection, and deterministic tie-breaking algorithms to derive authoritative execution metrics with $\pm 1.0s$ wall-clock accuracy.
- Packaged the engine into a sandboxed execution runtime with deterministic PyTest test fixtures verifying edge-case DAG topologies.

TECHNICAL SKILLS

- **Programming Languages:** Python (PyTest, NumPy, Pandas, AsyncIO), C/C++, Bash Scripting, SQL (PostgreSQL, SQLite), OpenCL C
- **Systems & Infrastructure:** Linux (Storage & Block Layers, LVM, Kernel Internals, Administration), Docker (Multi-stage, Containerization), Git & GitHub, CI/CD Pipelines
- **Core Computer Science:** Data Structures & Algorithms (DSA), High-Performance Computing (SIMD/SIMT), Distributed Systems, Test-Driven Development (TDD)
- **Developer Tooling:** PyTest, GCC/Clang, Make, GDB, SQLite3, Thin-Provisioning Tools, ReportLab

EDUCATION & CERTIFICATIONS

Bachelor of Technology (B.Tech), Computer Science & Engineering

July 2024 – June 2028 (Expected)

Institute of Advanced Research (IAR) — Gandhinagar, India

- **Google Cybersecurity Professional Certificate** — Coursera / Google (Python Automation, Linux, SQL, Network Security)
- **Google Cloud Fundamentals: Core Infrastructure** (Google Cloud) | **Introduction to Networking** (NVIDIA) | **Git & GitHub** (IBM)
- **Industry Job Simulations:** AWS (Solutions Architecture), Accenture Nordics (Software Engineering), Mastercard (Technology)